

Snehit Chunarkar

ML Researcher | Noise Robust Speech Emotion

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Summary

ML Researcher focused on noise-robust speech emotion recognition and multimodal audio-text learning | PhD at NTHU's BIIC Lab under Prof. Chi-Chun Lee (Jeremy).

Research focus: speech/audio representation learning, emotion understanding, and cross-modal fusion (audio-text), with an emphasis on robust modeling and interpretability.

Experience with top-tier models: WavLM, Wav2Vec2, CLAP, Whisper, RoBERTa. Globally #2 in ICASSP 2026 SP Grand Challenge (XACLE) with 92% better prediction; led a noise-robust speech emotion recognition system with 32% improvement.

Education

PhD, Electrical Engineering (2027 expected)

[National Tsing Hua University \(NTHU\)](#), Hsinchu, Taiwan

M.Tech., Instrumentation and Signal Processing

[Indian Institute of Technology \(IIT\)](#), Roorkee, India

B.E., Instrumentation Engineering

[Government College of Engineering](#), Chandrapur, India

Research Advisor

Prof. Chi-Chun Lee (Jeremy)

Professor & Associate Chair, NTHU; Research Fellow, Academia Sinica; Director, NVIDIA-NTHU Joint Innovation Center

Publications

Reasoning Driven Captions to Assist Noise Robust Speech Emotion Recognition

Snehit B. Chunarkar, Chi-Chun Lee - IEEE ICASSP 2026 (Barcelona, Spain)

Cross-Modal Semantic Alignment Via Ensemble Audio-Text Features for XACLE Challenge

Snehit B. Chunarkar, Krishnagiri Hamza, Chi-Chun Lee - IEEE ICASSP Workshop 2026 (Barcelona, Spain)

Structured Event-to-Text Generation for Zero-Shot Audio Description: A Prompt Study Across LLM Sizes

Snehit B. Chunarkar, Chi-Chun Lee - Preprint: ResearchGate (Under Review: DCASE)

Exploring Pretrained Audio-Text Encoders for Audio Moment Retrieval: DCASE 2026 Task 6

Snehit B. Chunarkar, Krishnagiri Hamza, Chi-Chun Lee - DCASE2026 Challenge

STELIN-US: A Spatio-Temporally Linked Neighborhood Urban Sound Database

S. Chunarkar, B. Su, C. Lee - DCASE2023 (Tampere, Finland)

Mixed Language Separation Using Deep Neural Network

S. Chunarkar, S. Chiluveru, M. Tripathy - ICEECCOT 2021, (Mysuru, India)

Efficient Hardware Implementation of Nonlinear Activation Function for Inference Model

S. Chunarkar, S. Chiluveru - ISOC 2024, (Sapporo, Japan)

Projects

Reasoning-driven-SER: (Speech Emotion Recognition in Noisy Environment)

ICASSP 2026 / Most Competitive year with only 40% acceptance rate

Integrating Reasoning-enriched caption and Audio using Cross-Attention to enhance Speech Emotion Recognition (SER) in Noisy Environment.

Improved SER by 32% over the conventional Audio-Only approach at SNR: -10dB.

EnsembleSVR-XACLE: (Audio-Text Alignment Prediction)

ICASSP 2026 SP Grand Challenge: XACLE / Rank #2 (Globally)

Engineered end-to-end pipeline to predict audio-text alignment scores in multimodal datasets

Ranked #2 on official [leaderboard](#) among 40+ submissions

SED2Text: (Sound Events in Natural Language)

[Preprint] ResearchGate / [Under Review] DCASE 2026

A training-free, 0.59 billion-parameter cascade for temporally grounded audio description.

My proposed approach can replace large audio-language models for this task.

AMR Encoder Exploration: (Audio Moment Retrieval)

DCASE 2026 Challenge - Task 6: XACLE / Top 12 (Globally)

Explored Audio-Text Encoders for Audio Moment Retrieval (AMR)

116.66% improvement in Recall@0.7 on Evaluation data compared to Baseline - [leaderboard](#).

STeLiN-US Database (Dataset synthesis for real-world SED)

DCASE Publication

Synthesized a dataset to mimic real-world acoustic environments (overlapping events, spatial distance scaling, temporal correlation) for training sound event detection systems.

Pawpularity Contest: (Cuteness Score Prediction for Pet's Image)

NTHU Machine Learning Course Project (11230EE 655000) / Rank #2 in class

Built ensemble regression model (Gradient Boosting, SVR, Neural Networks) to predict pet adoption likelihood from image and metadata features

Ranked #2 among 16 course teams; with state-of-the-art performance on Pawpularity dataset

Skills

Programming Languages: Python (95%), Bash (50%), Markdown (85%)

Machine Learning & Deep Learning: PyTorch, NumPy, Pandas, Matplotlib, scikit-learn, Hugging Face Transformers

Specialized Tools: Audio processing (librosa, pydub), Signal processing (SciPy), GPU computing (CUDA, cuML), Dataset platforms (Kaggle, Zenodo)

Development Environments: VS Code, Spyder, Jupyter Notebook, MATLAB, MobaXterm

Cloud & HPC: Google Colab, TWCC (Taiwan Computing Cloud)

Additional Information

Languages: English (fluent), Hindi (native), Marathi (native)

Active Engagement: Open-source GitHub, Zenodo contributions